Homework 8

William Flanders*
Haverford College
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1. NEWMAN EXERCISE 6.4

We look to model the circuit of resistors in Figure 1. The voltage source, V_+ provides 5 V. We can utilize Ohm's Law and Kirchoff current law to find the voltages at each junction. The result is a system of four equations. For the junction at V_1,

$$4V_1 - V_2 - V_3 - V_4 = V_+ \tag{1}$$

for the junction at V_2,

$$V_1 - 3V_2 + V_4 = 0 \tag{2}$$

for the junction at V_3,

$$V_1 - 3V_3 + V_4 = -V_+ \tag{3}$$

and for the junction at V_4,

$$V_1 + V_2 + V_3 - 4V_4 = 0. \tag{4}$$

We can combine these four equations into a matrix equation,

$$\begin{bmatrix} 4 & -1 & -1 & -1 \\ 1 & -3 & 0 & 1 \\ 1 & 0 & -3 & 1 \\ 1 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} V_+ \\ 0 \\ -V_+ \\ 0 \end{bmatrix} \tag{5}$$

where we can now employ linear algebra techniques to solve. We can use row reduction techniques to perform LU decomposition. This forms an upper triangular matrix that allows us to use backward substitution to solve for the unknown voltage values. We will create a python program with the numpy linear algebra solve method that employs this technique.

We begin with pseudo code:

```
import dependencies

initailize coefficient matrix
initialize ordinate values

np.linalg.solve

return values of V
```

We find that $V_1 = 3$ V, $V_2 = 1.67$ V, $V_3 = 3.33$ V, and $V_4 = 2$ V.

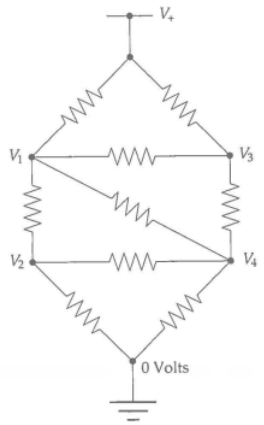

FIG. 1: Circuit of resistors.

2. NEWMAN EXERCISE 6.9

We look to model an asymmetric quantum well. Suppose we have a particle of mass M in a one-dimensional quantum well of width L that isn't square. Let's also assume that the potential varies within the well. We note that the spatial part $\psi(x)$ of the wavefunction is an eigenfunction that obeys the time-independent Schrodinger equation $\hat{H}\psi(x) = E\psi(x)$, where the Hamiltonian operator $\hat{H}$ is given by

$$\hat{H} = \frac{-\hbar^2}{2M}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x). \tag{6}$$

Assuming the walls are infinitely high and the wavefunction is zero outside the well, the wavefunction can be expressed as a Fourier sine series,

$$\psi(x) = \sum_{n=1}^{\infty} \psi_n \sin\frac{\pi n x}{L} \tag{7}$$

where ψ_1, psi_2, ... are the Fourier coefficients. Starting with

$$\int_0^L \sin\frac{\pi m x}{L}\sin\frac{\pi n x}{L}\,dx = \begin{cases} \frac{L}{2} & \text{if } m = n \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

we can rewrite this expression such that

$$\int_0^L \psi_m(x)\psi_n(x)\,dx = \frac{L}{2}\delta_{mn} \tag{9}$$

*Electronic address: wflanders@haverford.edu; URL:

where δ_{mn} is the Kronecker delta and is defined by

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n. \end{cases} \tag{10}$$

Now, we want to consider

$$\sum_{n=1}^{\infty} \psi_n \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} \, dx. \tag{11}$$

We can use the time-independent Schrodinger equation to substitute $E\psi_n$ for $\hat{H}\psi_n$ and the definitions of ψ_n and ψ_m to yield

$$\sum_{n=1}^{\infty} \psi_n E \int_0^L \psi_m(x)\psi_n(x) \, dx. \tag{12}$$

We know the value of this integral from Equation 9,

$$\sum_{n=1}^{\infty} \psi_n E \frac{L}{2} \delta_{mn}. \tag{13}$$

Finally, the definition of the Kronecker delta implies that this solution is non-trivial only when $n = m$, so as we sum across all ψ_n, only when we have ψ_m will the solution be non-zero. This shows that

$$\sum_{n=1}^{\infty} \psi_n \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} \, dx = \frac{1}{2} L E \psi_m \tag{14}$$

We can construct a Hamiltonian matrix $\mathbf{H}$ with entries

$$\begin{aligned} H_{mn} &= \frac{2}{L} \int_0^L \sin \frac{\pi m x}{L} \hat{H} \sin \frac{\pi n x}{L} \, dx \\ &= \frac{2}{L} \int_0^L \sin \frac{\pi m x}{L} \left(\frac{-\hbar^2}{2M} \frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x) \right) \sin \frac{\pi n x}{L} \, dx. \end{aligned} \tag{15}$$

We will want to consider the case when $V(x) = ax/L$. Plugging this into Equation 15,

$$\begin{aligned} H_{mn} = \frac{n^2 \pi^2 \hbar^2}{M L^3} \int_0^L \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} \, dx \\ + \frac{2a}{L} \int_0^L x \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} \, dx \end{aligned} \tag{16}$$

Note that the second integral

$$\int_0^L x \sin \frac{\pi m x}{L} \sin \frac{\pi n x}{L} \, dx \tag{17}$$

evaluates to

$$= \begin{cases} L^2/4 & \text{if } m = n. \\ -(\frac{2L}{\pi})^2 \frac{mn}{(m^2-n^2)^2} & \text{if } m \neq n \text{ and one is even, one is odd} \\ 0 & \text{if } m \neq n \text{ and are both even or odd} \end{cases} \tag{18}$$

Equation 16 then simplifies to

$$\begin{cases} \frac{n^2 \pi^2 \hbar^2}{2ML^2} + \frac{aL}{2} & \text{if } m = n \\ -(\frac{8aL}{\pi^2}) \frac{mn}{(m^2-n^2)^2} & \text{if } m \neq n \text{ and one is even, one is odd} \\ 0 & \text{if } m \neq n \text{ and are both odd or even} \end{cases} \tag{19}$$

We can create a python program that creates the matrix $\mathbf{H}$ with these defined elements. We will consider $L = 5$Å, $a = 10$eV, $q_e = 1.6022 \cdot 10^{-19}$J, and $M_e = 9.1094 \cdot 10^{-31}$kg. We will look to create an NxN matrix for $\mathbf{H}$ and compute the corresponding energy eigenvalues. We will also look to plot the probability density functions for the first three energy eigenstates for range of $0 \leq x \leq L$.

We begin with pseudo code

```
import dependencies

initialize constants

def H(m,n)
    if m = n:
        first case
    elif m != n and one odd and one even:
        second case
    else:
        third case

for i in range(N):
    for j in range(N):
        fill out H matrix

use numpy to solve for eigenvalues of H

use numpy to solve for eigenvectors of H

def psi(n, x):
    calculate wavefunction

plot probability densities
```

For a 10x10 matrix, we find that the ten energy eigenvalues are 5.83639118, 11.18113355, 18.66298345, 29.14436203, 42.65533201, 59.18562839, 78.72986474, 101.28614293, 126.85221998, and 155.55635795 eV.

For a 100x100 matrix, we find that the first ten energy eigenvalues are 5.83639077, 11.1811322, 18.6629816, 29.1443532, 42.6553229, 59.1855758, 78.7298129, 101.285512, 126.851388, and 155.426736 eV.

This seems to suggest that regardless of $\mathbf{H}$'s size, the accuracy in the value of each eigenvalue changes only marginally.

Figure 2 shows the probability densities for the ground, first excited, and second excited states.

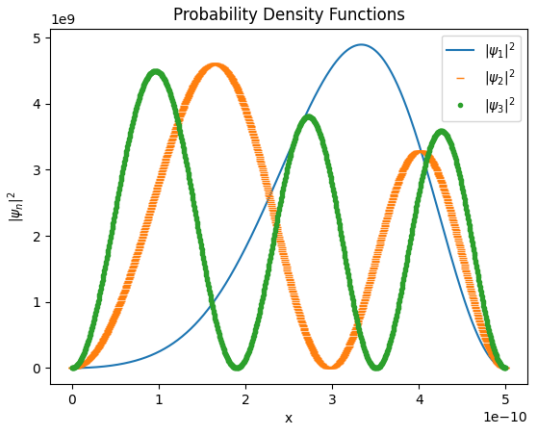

FIG. 2: Probability density functions for the ground, first excited, and second excited states.

From the plots, we can see how a non-zero potential in an asymmetric well causes our expectation of where the electron is to change compared to a square well. In a square well at the ground state, we'd expect it to be in the center. In this asymmetric well, we expect to the electron to be located more towards the right. Interestingly, as the electron's energy state increases, our expectation moves to the left significantly.

3. SURVEY

This problem set took roughly 5-6 hours. Exercise 6.4 was a good chance to practice matrix operations within python. Exercise 6.9 was somewhat tedious with the latex typesetting and having to show the mathematics, however, was satisfying to complete.